

RGTVetrex — AI Lead Outbound Engine

Product Overview & Technical Documentation

What is this system?

Ek fully automated AI-powered outbound sales engine jo aapke liye automatically leads dhundta hai, qualify karta hai, personalized emails likhta hai, aur send karta hai — bina kisi manual kaam ke.

Simple words mein: Aap bas batao "mujhe Mumbai ke SaaS startups ke founders chahiye" — system baaki sab khud karta hai.

Complete Pipeline Flow



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STAGE 4 – AI EXTRACTION

Gemini/Llama reads scraped content

Extracts: Name, Email, Phone, Title, Company

Fixes obfuscated emails: "name at domain dot com"

Validates email format

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STAGE 5 – DEDUPLICATION & VALIDATION

→ Root domain dedup (same company = 1 lead)

→ Email format validation

→ Neo4j graph database check (already contacted?)

→ Best contact selection (named person > generic)

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STAGE 6 – AI QUALIFIER

LLM reviews and standardizes each lead

Confidence scoring (0.0 → 1.0)

Drops only malformed/no-contact-info leads

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STAGE 7 – HUMAN APPROVAL GATE ← CLIENT REVIEWS

Dashboard shows all qualified leads

Client approves / rejects individual leads

AI pre-generates personalized email draft

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STAGE 8 – PERSONALIZED EMAIL DRAFTING

Llama 70B writes hyper-personalized cold email

Uses company description for personalization

RAG system pulls successful past templates

Under 100 words, professional tone

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STAGE 9 – AUTOMATED EMAIL SENDING

Sends via configured SMTP

2-second rate limit between emails

Logs every send to database

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STAGE 10 – REPLY MONITORING & FOLLOW-UPS

Monitors inbox for replies

Auto-schedules follow-up sequences

Tracks: sent / replied / bounced / positive

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STAGE 11 – DASHBOARD & ANALYTICS

Live campaign status

Lead growth charts

Industry distribution

Success rate metrics

Tech Stack

Component	Technology	Purpose
Backend	FastAPI (Python)	REST API, campaign management
Frontend	React + TypeScript + Vite	Dashboard UI
Primary LLM	NVIDIA Llama 3.1 70B	Extraction, drafting
Fallback LLMs	Groq, OpenRouter, Gemini 2.5 Flash	Reliability chain
Graph DB	Neo4j Aura	Lead memory, deduplication
Vector DB	Qdrant Cloud	Email template RAG
SQL DB	Supabase (PostgreSQL)	Users, campaigns, tasks, notes
Scraping Tier 1	Tinyfish Fetch API	Cloud-based bypass scraper
Scraping Tier 2	curl_cffi	TLS/JA3 fingerprint spoofing
Scraping Tier 3	Camoufox	Anti-detect Firefox browser
Scraping Tier 4	Playwright + Stealth	Headless Chromium + stealth patches
Scraping Tier 5	BeautifulSoup (7-layer)	Static HTML deep extraction
Search	Tinyfish Search + DDG	URL discovery
Email	SMTP (Gmail)	Outbound email sending

Scraper — Full Technical Details

5-Tier Anti-Bot Strategy

Tier 1 — Tinyfish Fetch API

- Cloud-based fetcher with built-in bot bypass
- Returns clean markdown — no HTML parsing needed
- Fastest tier (~1-2 seconds per page)

Tier 2 — curl_cffi (TLS/JA3 Fingerprint Spoof)

- Python's default `requests` library has a unique TLS handshake that Cloudflare detects instantly

- `curl_cffi` compiles low-level curl code that sends Chrome 120/124/131 or Safari 17/18 TLS fingerprints
- Sites see a real browser at the TCP network level
- Bypasses ~60% of bot blocks without launching any browser
- Profiles rotated: `chrome120`, `chrome124`, `chrome131`, `safari17_0`, `safari18_0`

Tier 3 — Camoufox (Anti-Detect Firefox)

- Custom Firefox build with stealth patches at the engine level
- Removes automation indicators (`navigator.webdriver`, canvas fingerprint, WebGL leaks)
- Handles Cloudflare Turnstile and complex JS-rendered pages
- Best for Indian government/education websites

Tier 4 — Playwright + Stealth

- Full headless Chromium browser
- Stealth JS injected before every page load:
 - Removes `navigator.webdriver = true` flag
 - Fakes `navigator.plugins` (5 plugins)
 - Fakes `navigator.languages`
 - Adds `window.chrome` object
- **Bézier mouse paths** — curved organic mouse movement, not straight lines
- **Poisson-distributed delays** — random 1-7 second waits, not fixed `sleep(2)`
- Heavy assets (images, fonts, videos) blocked to save bandwidth

Tier 5 — BeautifulSoup 7-Layer Extraction

1. `mailto:` and `tel:` links → direct email/phone extraction
2. Meta tags (OG, Twitter Card, schema.org, author)
3. JSON-LD structured data (`<script type="application/ld+json">`)
4. Microdata (`itemprop` attributes — contact schemas)
5. HTML comments containing `@` or contact keywords
6. `<address>` HTML tags (spec-defined contact blocks)
7. Visible text keyword scan (email, phone, contact, director etc.)

Session Cookie Cache

- After Camoufox/Playwright solves a Cloudflare challenge, cookies are cached for 30 minutes
- Subsequent pages on the same domain use cached cookies with `curl_cffi` (fast path)
- Only falls back to browser if cookies expire
- **Result:** First visit = 5-8 seconds, subsequent pages = 1-2 seconds

Hidden JSON API Detection

- Before scraping any site, checks 10 common API paths:
 - `/api/contacts`, `/api/staff`, `/api/team`
 - `/wp-json/wp/v2/users`
 - `/api/members`, `/.json`, `/data.json`

- If site returns JSON directly → 10x faster, 100% accurate, no HTML parsing needed

Lead Sources

Source	Method	What You Get
Company Websites	5-tier scraper	Name, email, phone, title
LinkedIn X-Ray	<code>site:linkedin.com/in</code> via DDG	Name, title, company, LinkedIn URL, email (if public)
Google Dorking	DDG + LLM parsing	Emails from social bios, PDFs, contact pages
Reddit	PRAW official API	Business owners who posted contact info
Quora	Playwright scraping	Professionals who listed email in bio/answers
Instagram	<code>site:instagram.com</code> <code>"@gmail.com"</code> dork	Public email from bio
PDF Directories	<code>filetype:pdf</code> dork	Staff lists, faculty directories

What We've Improved (Before vs After)

Feature	Before	After
Scraper tiers	1 (requests only)	5 (Tinyfish → curl_cffi → Camoufox → Playwright → BS4)
User agents	1 fixed	5 rotating
Retries	1 attempt	3 with Poisson backoff
BS4 extraction	Basic cleanup	7-layer deep extraction
Processing	Sequential (1 site at a time)	3 parallel workers
Duplicate detection	URL string only	Root domain dedup
Same company leads	Multiple counted	1 best contact kept
LLM extraction	<code>with_structured_output</code> (silent fails)	Manual JSON parsing
Query generation	Always "B2B startup" framing	Respects exact user input
Email validation	Loose regex (caught filenames)	Strict + throwaway filter
Obfuscated emails	Not handled	Fixed automatically

Feature	Before	After
Lead sources	Company websites only	7 sources
Blocked sites (403)	Failed permanently	Camoufox/curl_cffi bypass
JS-rendered pages	Empty result	Playwright renders JS

LinkedIn — Honest Status

What you get via X-Ray:

- ☒ Full Name
- ☒ Job Title
- ☒ Company Name
- ☒ LinkedIn Profile URL
- ☐ Email — only if publicly listed (~10-15% of profiles)

Why direct LinkedIn scraping is not possible:

- LinkedIn uses behavioral analysis + IP reputation + login walls
- Even paid tools like PhantomBuster get banned regularly
- The X-Ray approach searches Google's cached index of public profiles
- Zero LinkedIn contact → zero ban risk

Workaround for missing emails: When email is not on LinkedIn profile, the system uses the company name to scrape the company's official website and find the email there — which happens automatically in Stage 2.

Key Differentiators vs Competitors

vs Apollo.io / Hunter.io / ZoomInfo

Feature	Apollo.io	RGTVetrex
Cost per lead	\$0.10 - \$0.50	Near zero (API costs only)
Email sending	Separate tool needed	Built-in
Personalization	Template-based	AI-written per person
Human approval	Not available	Full approval gate
Lead memory	Basic CRM	Neo4j graph (permanent)
Custom sources	No	Reddit, Quora, PDFs, X-Ray

vs Manual Outreach

Metric	Manual	RGTVetrex
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Metric	Manual	RGTVetrex
Time per day	2-3 hours	10 minutes (just review)
Leads per day	20-30	Up to 100
Duplicates	Common	Zero (Neo4j memory)
Email quality	Variable	AI-personalized always
Follow-ups	Easy to forget	Auto-scheduled

ROI Summary

- **Setup time:** 10 minutes per campaign
- **Review time:** 5-10 minutes (human approval gate)
- **Cost:** API costs only (LLMs mostly free tier)
- **Output:** Up to 100 qualified leads/day with personalized emails ready to send
- **No per-lead pricing** — unlike Apollo (\$0.10-0.50/lead), costs stay flat regardless of volume

Running the System

Backend:

```
cd lead-outbound-engine
venv\Scripts\activate
uvicorn api.main:app --reload --host 0.0.0.0 --port 8000
```

Frontend:

```
cd lead-outbound-engine\frontend-web
npm run dev
```

URLs:

- Backend API: <http://localhost:8000>
- Frontend Dashboard: <http://localhost:5173>
- API Docs: <http://localhost:8000/docs>